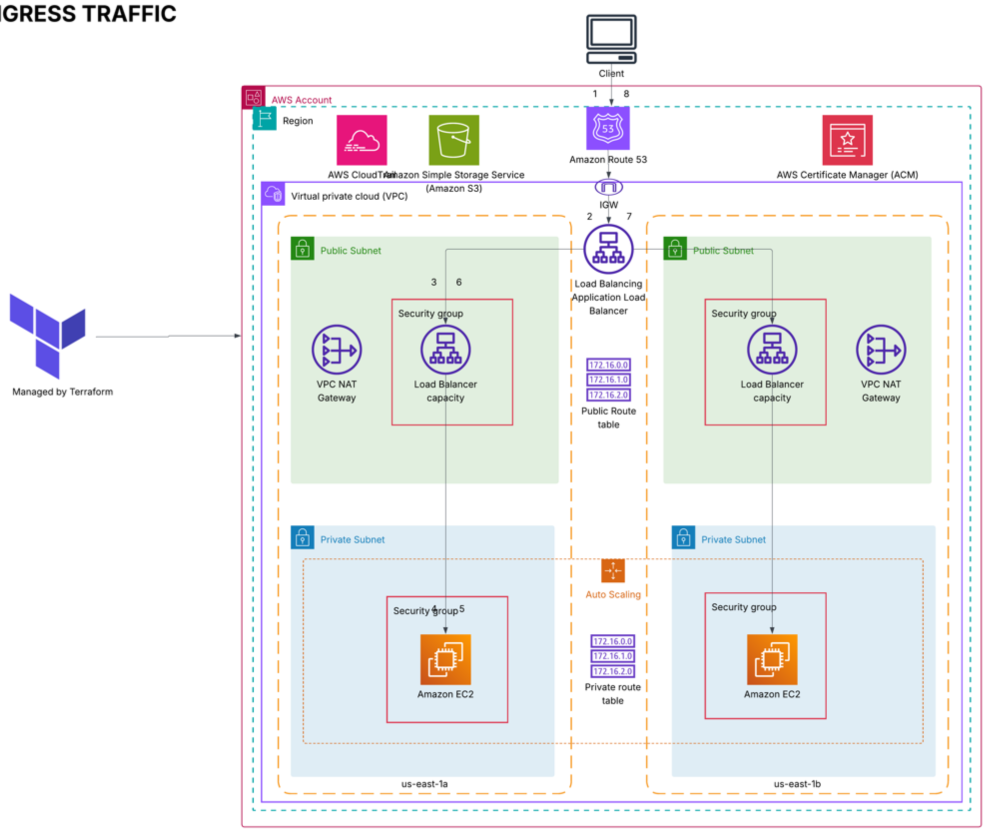
Traffic flow for Ingress and Egress

INGRESS TRAFFIC



Ingress Traffic Summary

1. The client sends a request to the application domain.
2. Route 53 resolves the domain name to the Application Load Balancer (ALB).
3. The request enters the VPC through the Internet Gateway (IGW).
4. The ALB receives the request and terminates SSL/TLS using the ACM certificate.
5. The ALB forwards the request to the Target Group.
6. The Target Group routes the request to a healthy EC2 instance in the private subnet.
7. The EC2 instance processes the request and returns the response.
8. The response is sent back through the ALB to the client.

Ingress Traffic Flow

Client
↓
Route 53

↓
Application Load Balancer (ALB)
↓
Target Group
↓
EC2 Instance (Private Subnet)
↓
Application Response
↓
ALB
↓
Client

Egress Traffic Summary

1. EC2 instance in the private subnet initiates an outbound request.
2. The private route table forwards internet-bound traffic (0.0.0.0/0) to the NAT Gateway.
3. The NAT Gateway performs source NAT using its Elastic IP.
4. Traffic is routed through the public route table to the Internet Gateway.
5. The Internet Gateway forwards the traffic to the destination on the internet.
6. Return traffic follows the reverse path through the Internet Gateway and NAT Gateway back to the originating EC2 instance.

Engress Traffic Flow

EC2 Instance (Private Subnet)

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Private Route Table

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NAT Gateway

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Public Route Table

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Internet Gateway (IGW)

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Internet / External Service

↓

Internet Gateway (IGW)

↓

NAT Gateway

↓

EC2 Instance